

ELIZABETH MUNCH

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Dept of Computational Mathematics, Science, and Engineering $\diamond$ Dept of Mathematics
Michigan State University $\diamond$ East Lansing, MI

EDUCATION

Duke University, *Durham, NC*

Ph.D. Dept. of Mathematics

May 2013

· Thesis: Applications of Persistent Homology to Time Varying Systems

M.A. Dept. of Mathematics

Dec 2010

University of Rochester, *Rochester, NY*

B.S. Mathematics, Summa Cum Laude, School of Arts and Sciences

May 2008

B.M. Harp Performance with High Distinction, Eastman School of Music

May 2008

RESEARCH EXPERIENCE

Michigan State University, *East Lansing, MI*

Dept. of Computational Mathematics, Science and Engineering (CMSE)

Dept. of Mathematics

Associate Professor

July 2022 – Present

Assistant Professor

Aug 2017 – July 2022

University at Albany – SUNY, *Albany, NY*

Dept. of Mathematics & Statistics

Assistant Professor

Sept 2014 – July 2017

Dept. of Computer Science

Affiliated Faculty

July 2015 – July 2017

University of Minnesota, *Minneapolis, MN*

Institute for Mathematics and Its Applications

Postdoctoral Fellow

Sept 2013 – Aug 2014

Duke University, *Durham, NC*

Dept. of Mathematics

Visiting Assistant Professor

June 2013 – Aug 2013

Graduate Research Assistant

Sept 2008 – May 2013

AWARDS AND HONORS

Withrow Teaching Excellence Award, College of Engineering, Michigan State University, 2023

NSF CAREER Award, 2022

Jo Rae Wright Fellowship for Outstanding Women in Science, Duke University, 2012-2013

Phi Beta Kappa, University of Rochester, May 2008

Performer's Certificate in Harp, May 2008

Lois S. Rogers Scholarship, Eastman School of Music, 2004-2008

Performer's Certificate, Eastman School of Music, 2008

Presser Scholarship, Eastman School of Music, 2007

SELECT PUBLICATIONS

- [1] Erin Wolf Chambers, Elizabeth Munch, Sarah Percival, and Bei Wang. “Bounding the interleaving distance for mapper graphs with a loss function”. In: *Journal of Applied and Computational Topology* 9.3 (July 2025). ISSN: 2367-1734. DOI: [10.1007/s41468-025-00215-x](https://doi.org/10.1007/s41468-025-00215-x). arXiv: [2307.15130](https://arxiv.org/abs/2307.15130) [cs.CG].
- [2] Elizabeth Munch. “An Invitation to the Euler Characteristic Transform”. In: *The American Mathematical Monthly* 132.1 (2025), pp. 15–25. DOI: [10.1080/00029890.2024.2409616](https://doi.org/10.1080/00029890.2024.2409616). arXiv: [2310.10395](https://arxiv.org/abs/2310.10395) [cs.CG].
- [3] Sunia Tanweer, Firas A. Khasawneh, Elizabeth Munch, and Joshua R. Tempelman. “A topological framework for identifying phenomenological bifurcations in stochastic dynamical systems”. In: *Nonlinear Dynamics* (Feb. 2024). ISSN: 1573-269X. DOI: [10.1007/s11071-024-09289-1](https://doi.org/10.1007/s11071-024-09289-1). arXiv: [2305.03118](https://arxiv.org/abs/2305.03118) [math.DS].
- [4] Audun Myers, David Muñoz, Firas A. Khasawneh, and Elizabeth Munch. “Temporal Network Analysis Using Zigzag Persistence”. In: *EPJ Data Science* 12.6 (2023). DOI: [10.1140/epjds/s13688-023-00379-5](https://doi.org/10.1140/epjds/s13688-023-00379-5). arXiv: [2205.11338](https://arxiv.org/abs/2205.11338) [cs.CG].
- [5] Audun D. Myers, Max M. Chumley, Firas A. Khasawneh, and Elizabeth Munch. “Persistent homology of coarse-grained state-space networks”. In: *Phys. Rev. E* 107 (3 2023), p. 034303. DOI: [10.1103/PhysRevE.107.034303](https://doi.org/10.1103/PhysRevE.107.034303). arXiv: [2206.02530](https://arxiv.org/abs/2206.02530) [stat.ML].
- [6] İsmail Güzel, Elizabeth Munch, and Firas A. Khasawneh. “Detecting bifurcations in dynamical systems with CROCKER plots”. In: *CHAOS* 32.093111 (2022). Full version of [Guezel2022]. Chosen as featured article in Chaos. DOI: [10.1063/5.0102421](https://doi.org/10.1063/5.0102421). arXiv: [2206.04861](https://arxiv.org/abs/2206.04861) [math.DS].
- [7] Jose A. Perea, Elizabeth Munch, and Firas A. Khasawneh. “Approximating Continuous Functions on Persistence Diagrams Using Template Functions”. In: *Foundations of Computational Mathematics* (2022). DOI: [10.1007/s10208-022-09567-7](https://doi.org/10.1007/s10208-022-09567-7). arXiv: [1902.07190](https://arxiv.org/abs/1902.07190) [cs.CG, math.AT, math.ST, stat.ML, stat.TH].
- [8] Erik J Amézquita, Michelle Y Quigley, Tim Ophelders, Jacob B. Landis, Daniel Koenig, Elizabeth Munch, and Daniel H Chitwood. “Measuring hidden phenotype: Quantifying the shape of barley seeds using the Euler Characteristic Transform”. In: *in silico Plants* (2021). DOI: [10.1093/insilicoplants/diab033](https://doi.org/10.1093/insilicoplants/diab033). bioRxiv: [10.1101/2021.03.27.437348](https://doi.org/10.1101/2021.03.27.437348).
- [9] Adam Brown, Omer Bobrowski, Elizabeth Munch, and Bei Wang. “Probabilistic convergence and stability of random mapper graphs”. In: *Journal of Applied and Computational Topology* (2021). DOI: [10.1007/s41468-020-00063-x](https://doi.org/10.1007/s41468-020-00063-x). arXiv: [1909.03488v1](https://arxiv.org/abs/1909.03488v1).
- [10] Sarah Tymochko, Elizabeth Munch, Jason Dunion, Kristen Corbosiero, and Ryan Torn. “Using Persistent Homology to Quantify a Diurnal Cycle in Hurricane Felix”. In: *Pattern Recognition Letters* 133 (2020), pp. 137–143. ISSN: 0167-8655. DOI: [10.1016/j.patrec.2020.02.022](https://doi.org/10.1016/j.patrec.2020.02.022). arXiv: [1902.06202](https://arxiv.org/abs/1902.06202).
- [11] Audun Myers, Elizabeth Munch, and Firas A. Khasawneh. “Persistent Homology of Complex Networks for Dynamic State Detection”. In: *Physical Review E* 100.2 (2019), p. 022314. DOI: [10.1103/PhysRevE.100.022314](https://doi.org/10.1103/PhysRevE.100.022314). arXiv: [1904.07403](https://arxiv.org/abs/1904.07403).
- [12] Vin de Silva, Elizabeth Munch, and Anastasios Stefanou. “Theory of interleavings on categories with a flow”. In: *Theory and Applications of Categories* 33.21 (2018), pp. 583–607. arXiv: [1706.04095](https://arxiv.org/abs/1706.04095).
- [13] Firas A. Khasawneh and Elizabeth Munch. “Topological data analysis for true step detection in periodic piecewise constant signals”. In: *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Science* 474.2218 (2018), p. 20180027. DOI: [10.1098/rspa.2018.0027](https://doi.org/10.1098/rspa.2018.0027). arXiv: [1805.06403](https://arxiv.org/abs/1805.06403).
- [14] Elizabeth Munch. “A User’s Guide to Topological Data Analysis”. In: *Journal of Learning Analytics* 4.2 (2017). DOI: [10.18608/jla.2017.42.6](https://doi.org/10.18608/jla.2017.42.6).
- [15] Vin de Silva, Elizabeth Munch, and Amit Patel. “Categorified Reeb Graphs”. In: *Discrete & Computational Geometry* (2016), pp. 1–53. ISSN: 1432-0444. DOI: [10.1007/s00454-016-9763-9](https://doi.org/10.1007/s00454-016-9763-9). arXiv: [1501.04147](https://arxiv.org/abs/1501.04147).
- [16] Firas A. Khasawneh and Elizabeth Munch. “Chatter detection in turning using persistent homology”. In: *Mechanical Systems and Signal Processing* 70-71 (2016), pp. 527–541. ISSN: 0888-3270. DOI: [10.1016/j.ymssp.2015.09.046](https://doi.org/10.1016/j.ymssp.2015.09.046).

SELECT CURRENT GRANTS

- **CAREER: Reeb graph learning: Classification, Clustering, and Embedding of Graphical Signatures.** NSF [CCF-2142713](#): \$507,462
- **Collaborative Research: AF: Medium: A Unified Framework for Geometric and Topological Signature-Based Shape Comparison.** NSF [CCF-2106578](#): \$409,945
- **Collaborative Research: RESEARCH-PGR: Predicting Phenotype from Molecular Profiles with Deep Learning.** NSF [IOS-2310355](#): \$639,998
- **Topological Signal Processing of Stochastic Dynamical Systems.** AFOSR FA9550-26-1-0011: \$393,310
- **Parameter Space Navigation Using Topological Optimization.** US Army Research Office: \$300,000
- **NRT-AI-HDR: Harnessing the data revolution to enable predictive multi-scale modeling across STEM.** NSF [DGE-2152014](#): \$2,968,065

SELECT RECENT TALKS

1. **Plenary**, Topological methods for time-varying data: theory and applications (Top Time), ANU, Canberra, Australia, October 27, 2025.
2. Workshop, BIRS, Banff, Canada, August 12, 2025. [Recording available](#)
3. **Lecture Series**, EWM-EMS Summer School: Stability in topological data analysis, Institut Mittag-Leffler, Stockholm, Sweden., June 30 - July 4, 2025. [Materials available](#)
4. **Plenary**, SIAM Dynamical Systems, Denver, CO., May 12, 2025.
5. **Semi-Plenary**, 58th Spring Topology and Dynamics Conference, Christopher Newport University, Newport News, VA, March 8, 2025.
6. Math Department Colloquium, Purdue University, West Lafayette, IN, Dec 11, 2024.
7. Workshop, Schloss Dagstuhl, Wadern, Germany, Feb 27, 2024.
8. CSCS Colloquium, University of Michigan, Ann Arbor, MI, Feb 8, 2024.
9. ACMS Colloquium, Notre Dame, South Bend, IN, Jan 11, 2024.
10. **Plenary**, SIAM Great Lakes Meeting, Michigan State University, East Lansing, MI, Oct 14, 2023.
11. Computational Persistence Workshop, Purdue, West Lafayette, IN, Sep 27, 2023. [Recording available](#)
12. ICIAM, Tokyo, Japan (Virtual), Aug 25, 2023.
13. TDA Week, Kyoto University, Kyoto, Japan, Aug 1, 2023.
14. UCLA Math Department Colloquium, Feb 23, 2023.
15. SIAM-MDS, San Diego, CA, Sep 29, 2022.
16. **Plenary talk**, ATMCS 10, Oxford, UK, June 23, 2022.
17. **Keynote Address**, WiDS Iowa, Virtual, Apr 21, 2022.
18. Math Dept Colloquium, University of Utah, Salt Lake City, UT, Apr 7, 2022.
19. **Keynote Address**, NAPPN Annual Conference, University of Georgia, Athens, GA, Twosday, Feb 22, 2022.
20. Mathematical Congress of the Americas, Online due to COVID-19, July 16, 2021.

TEACHING EXPERIENCE

Course	Title	Institution	Semesters
CMSE 201	Intro to Computational Modeling and Data Analysis	MSU	Spring, 2018
CMSE 381	Fundamentals of Data Science Methods	MSU	Spring, 2022 Fall, 2022 Fall, 2023 Fall, 2024
CMSE 382	Optimization Methods in Data Science	MSU	Spring, 2026
CMSE 801	Intro to Computational Modeling	MSU	Spring, 2024 Spring, 2020
CMSE 802	Methods Computational Modeling	MSU	Spring, 2021
CMSE 491/890	Topological Analysis of Large Datasets	MSU	Fall, 2017 Fall, 2021 Fall, 2025
MTH 132	Calculus I	MSU	Fall, 2020
MTH 481	Discrete Mathematics I	MSU	Fall, 2018
MTH 960	Algebraic Topology I	MSU	Fall, 2024
MTH 994	Topics: The Interleaving Distance	MSU	Spring, 2023
AMAT 840	Applied Topology	UAlbany	Spring, 2015
AMAT 587	Graph Theory	UAlbany	Fall, 2015
AMAT 540B	Topology II	UAlbany	Spring, 2017
AMAT 502	Modern Computing for Mathematicians	UAlbany	Spring, 2016
AMAT 363	Statistics	UAlbany	Fall, 2014 Fall, 2015 Spring, 2016 Spring, 2017
TIP	Mobius Strips, Klein Bottles, and Fractals: The Mathematics of Distortion	Duke TIP	July, 2012
Math 32L	Laboratory Calculus II	Duke	Spring, 2011
Math 31L	Laboratory Calculus I	Duke	Fall, 2009
Math 25L	Laboratory Calculus and Functions I	Duke	Fall, 2008

CURRENT RESEARCH GROUP MEMBERS

<i>Graduate Students</i>	<i>Postdocs</i>	<i>Former Graduate Students</i>	<i>Former Postdocs</i>
Rachel Roca	Nkechi Nnadi	Anastasios Stefanou	Sarah Percival
David Muñoz	Yemeen Ayub	Sarah Tymochko	Sourabh Palande
Astrid Olave		Erik Amézquita	Shelley Kandola
Ishika Ghosh		Christopher Potvin	Tim Ophelders
Kevin Woytowich		Danielle Barnes	Michelle Quigley
		Sarah McGuire	
		Xinyi (Elena) Wang	
		Christopher Sukhu	
		Mitchell Eithun	

SELECT SERVICE

Departmental and University Service

MSU TDA Seminar, Creator and Co-Organizer	Mar 2020 - Present
CMSE Advisory Committee (AdCom), MSU	2017-2018, 2020-2026
CMSE Graduate Curriculum Committee, MSU	2023-Present
College of Engineering Graduate Studies Committee, MSU	2018-2020, 2023-2024
Various Hiring Committees, MSU	
Math Undergraduate Committee, MSU	Sept 2021- Feb 2022
CNS Strategic Plan Integration Committee, MSU	Jan 2020 - Dec 2020
Math Dept. Graduate Committee, UAlbany	Sept 2016 - Aug 2017

Professional Service

Program Director, SIAM Activity Group on Data Science (SIAM-MDS)	2026
Workshop Committee Member, Symposium on Computational Geomtry (SoCG)	2020
Program Committee Member, Women+ Data Science Monthly Meetup , MSU	Fall 2020
Mentor, National Alliance for Doctoral Studies in the Mathematical Sciences	Fall 2020 - present
Steering Committee Member, ATMCS	Mar 2016 - May 2026
Steering Committee Member, Women in Computational Topology	Sep 2016 - Dec 2024
Organizer, The 4th Annual Minisymposium on Computational Topology, CG Week at the Symposium on Computational Geometry	Jun 2015

Community Outreach

Interviewed for <i>Meet a Mathematician</i> , Museum of Math (Virtual)	Feb 3, 2025
Panelist on Careers in Academia, Women in Science Conference, Notre Dame	Oct 2018
Interview for The Girls' Angle Bulletin, ISSN 2151-5700	Oct/Nov 2015
NYS Master Teacher Program Application Review Committee	Sept 2014
Volunteer for FEMMES Capstone Events	April 2011, April 2012

Editorial Boards & Conference Program Committees

Inaugural Editorial Board Member, Computing in Geometry and Topology (CGT)	2021-Present
Associate Editor, EPJ Data Science	2024-Present
Associate Editor, Journal of Applied and Computational Topology (APCT)	2024 - Present
Associate Editor, SIAM Journal on Mathematics of Data Science (SIMODS)	2026 - 2028
Guest Editor, La Mathematica	2023-2024
<i>Advances in Applied and Computational Topology</i> , WinCompTop Special Issue	

PROFESSIONAL AFFILIATIONS

AMS, SIAM, AWM, ACM, Math Alliance, NAM, AHS